Sahaj Bhadja

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Summary

Software engineer with 2+ years building healthcare-adjacent and AI-enabled applications using **Python**, **Java**, Azure, SQL Server, React, and REST APIs. Experienced across SDLC, QA validation, monitoring, automation, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Software Engineering, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, OOP, Software Engineering, Data Structures and Algorithms

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected a multi-tenant **Python** application using Azure SQL and Blob Storage for 500+ users, optimizing SQL queries, data models, and access controls to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built Azure-backed integration workflows processing 5,000+ monthly jobs with validation, reconciliation, monitoring, recovery, and CI/CD releases, achieving 99%+ reliability while supporting QA and production troubleshooting across environments.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Built **React** interfaces and Python Flask services handling 10,000+ daily requests, integrating generative AI, secure file ingestion, MySQL, REST APIs, and responsive workflows while maintaining stable peak performance.
- Developed a multithreaded processing pipeline for 1,000+ files with validation, progress monitoring, defect recovery, and user-visible status feedback, reducing turnaround 60% while supporting repeatable QA verification across staged environments.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a Python **machine-learning pipeline** processing 100,000+ trajectory records, improving prediction quality 18% through reproducible experiments, automated validation, performance tracking, structured defect analysis, and repeatable model configuration.
- Implemented data-processing and model-evaluation workflows with PyTorch, configuration management, and experiment monitoring, improving simulation stability 15% while supporting systematic debugging, issue resolution, and comparative analysis across complex scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered enterprise integration software with automated builds and QA validation across Windows, Linux, and macOS, maintaining 20+ unit and integration test suites through Jenkins, PyTest, and Mockito.
- Developed object-oriented services and SQL data pipelines using Glue, S3, caching, and pagination, improving response times 30%, processing efficiency 68%, and maintainability across integrated applications with reliable operations.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed **React**, JavaScript, and Node.js features across eight services, integrating WebSockets and REST APIs to synchronize more than two million records across responsive user and operational workflows.
- Automated testing, deployment, and application monitoring with Jenkins, Docker, Jest, Mocha, Selenium, Sentry, and Azure, cutting release time 50% while strengthening validation, observability, and production support across environments.

Skills

Languages & UI: Python, Java, JavaScript, TypeScript, SQL, HTML, CSS, React.js, Node.js, Flask, FastAPI
Data & APIs: SQL Server, Azure SQL, PostgreSQL, MySQL, MongoDB, REST APIs, JSON, database programming, data modeling, query optimization, ETL
AI & Automation: Generative AI, LLM systems, PyTorch, Transformers, prompt workflows, background jobs, process automation, machine learning, model evaluation
Azure & Delivery: Microsoft Azure, Blob Storage, Azure monitoring, Git, GitHub Actions, Jenkins, CI/CD, Docker, deployment validation, application monitoring
Engineering Practices: Agile, Scrum, SDLC, unit testing, integration testing, regression testing, troubleshooting, production support, code reviews, technical documentation

Projects

HMIS | Live | *Healthcare Workflows, TypeScript, REST APIs, Sentry*
- Built a healthcare workflow application with role-based access, scheduling, notifications, dashboards, and Sentry monitoring, reducing administrative workload 30% through validated functionality, reliable APIs, and documented operational processes.

Chat2SQL | Code | *Java, TypeScript, Docker, Maven, SQL*
- Built a cross-platform Java and TypeScript application supporting 10+ databases, schema-aware query generation, structured validation, automated testing, and Docker-based deployment across Windows, macOS, and Linux development environments.

Big-Brain | Live | *Generative AI, LLaMA-2, PostgreSQL, REST APIs*
- Built a generative-AI document platform processing 50,000+ records with secure access, semantic search, summarization, incremental indexing, and monitoring, reducing manual review time 60% through intelligent automation at scale.